

Reducing CO₂ Levels Using Genetic Algorithms

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Abstract

This project provides a solution framework for minimizing CO₂ emission levels in a national power grid through machine learning methods. The data is collected via Energinet’s public API, which provides real-time data about Denmark’s power system. A monotone gradient-boosted ensemble infers the total emission rate implied by a dispatch, XGBoost regressors forecast demand and renewable production over a four-hour horizon, and a genetic ensemble searches for the emission-minimizing trajectory subject to the national energy balance, ramp limits, and an outcome-space plausibility floor. The reduction is measured against the counterfactual in which dispatch is held fixed. A rolling-origin backtest of 225 origins across five expanding training windows finds that the emissions model ranks two ways of serving the same load with a slope of 0.80 against metered outcomes, and that the schedule the optimizer proposes is 35% cleaner than the dispatch that actually ran. After a ranking discount that accounts for imperfect delivery of predicted savings, the figure that the evidence supports is approximately 27%. Most of that reduction is a reallocation of cross-border flows rather than a reduction in domestic combustion.

1 Introduction

The data is collected via Energinet’s public API¹ which provides real-time data about Denmark’s power system. The dataset prepared for this research consists of one year of records aggregated onto the 15-minute grid used for imbalance settlement in Denmark, giving 34,600 observations between September 2025 and September 2026.

The main elements of this project are: a monotone gradient-boosted ensemble for inferring CO₂ levels, XGBoost regressors for predicting demand and renewable energy production, and genetic algorithms for finding the optimal energy distribution with respect to CO₂ emission minimization. The goal is to infer the resource combinations that minimize CO₂ emissions, depending on future demand and renewables production.

A live dashboard publishes a new four-hour schedule every thirty minutes.² Section 5 reports a rolling-origin backtest of that same pipeline against the dispatch that actually ran.

2 Data and the energy balance

The national energy balance is the identity

$$D = \underbrace{P_{\geq 100} + P_{< 100}}_{\text{generation}} + R + \sum_{i \in \mathcal{B}} x_i, \quad (1)$$

where R denotes renewable production (solar and wind), $P_{\geq 100}$ and $P_{< 100}$ denote energy produced by power plants with installed capacity greater or equal to 100 MW and less than 100 MW respectively, and $\mathcal{B}$ is the set of *cross-border* interconnectors: DK1-DE, DK1-NL, DK1-GB, DK1-NO, DK1-SE, DK2-DE, DK2-SE and Bornholm-SE. Positive exchange values are imports, negative ones are exports.

The DK1-DK2 link is deliberately absent from $\mathcal{B}$. It is the Great Belt Power Link, which joins the two Danish bidding zones; power flowing across it is internal and cannot change national

¹<https://www.energidataservice.dk/tso-electricity/PowerSystemRightNow>

²<https://cristianlinco.github.io/CO2-Re-evaluation/>

supply. Energinet excludes it from its own `Exchange.Sum` field for precisely this reason, which is verifiable directly from the API: on an arbitrary record the eight cross-border flows sum to 3481.49 MW and `Exchange.Sum` reports 3481.49 MW, with the DK1-DK2 value of -371.57 MW sitting outside the total. The link remains a decision variable, because it carries zonal information the emissions model uses, but it does not contribute to Equation (1).

3 Methodology

3.1 I. Inferring CO₂ levels

A regression model infers CO₂ levels taking into account 12 features: renewable energy production (solar and wind), energy produced by power plants with installed capacity greater or equal to 100 MW, energy produced by power plants with installed capacity less than 100 MW, and the nine energy exchanges between Denmark and other countries or areas. The hyperparameters are tuned using randomized search, and the model is evaluated through time-series cross-validation.

The target is total emissions, not intensity. Energinet publishes CO₂ as an intensity in g/kWh. Intensity is confounded: in Denmark, hours with high output from large thermal plants are also hours with high wind, so an unconstrained model of intensity learns that thermal generation predicts *low* emissions. The model here predicts the total emission rate

$$\Theta = \frac{\text{CO}_2 \text{ intensity} \times D}{1000} \quad [\text{t/h}], \quad (2)$$

from which intensity is recovered as $1000 \Theta / D$. Under consumption-based accounting every supply source carries a non-negative emission factor, so total emissions are monotonically non-decreasing in every source. That is imposed as a hard constraint on all eleven controllable features. Renewables are left unconstrained, since they are not a decision variable.

The estimator is an ensemble, and it reports its own reliability. The deployed estimator is a gradient-boosted ensemble of regression trees carrying that monotonicity. A companion depth-capped tree partitions the feature space and supplies, for any candidate, the number of training observations $n(\mathbf{x})$ behind its leaf and the dispersion $\sigma(\mathbf{x})$ of the target within it. A Mahalanobis distance $d^2(\mathbf{x})$ to the training distribution measures how far a candidate sits from anything ever observed.

A boosted ensemble of regression trees is unbounded below. Softplus is applied to the raw output so that predicted emissions cannot be negative; the transform is strictly increasing and therefore preserves every monotonic constraint, and it is the identity to within floating point across the range the data occupies. Separately, for the demand and renewable conditions of each horizon step the model looks up comparable hours in the training record and returns the 5th percentile of their metered emission rate. That quantity is the *plausibility floor*: the optimizer is penalized for proposing an outcome cleaner than anything those conditions have ever produced.

3.2 II. Forecasting demand and renewable production

The XGBoost regressors provide multi-step forecasting for energy demand and renewable production over the next 16 time points, that is four hours ahead at 15-minute resolution. These forecasts are important because the optimal power distribution has to match demand continuously; a persistent mismatch between supply and demand is what a system operator resolves through reserves, and the schedule the optimizer proposes must not create one. The distributable energy

whose allocation is optimized is the difference between forecasted demand and forecasted renewable energy, $D_{XG} - R_{XG}$.

Renewable production is not extrapolated from its own recent output. Wind four hours ahead is not recoverable from the last few observations. The model consumes Energinet’s own physical forecast and learns a correction on top of it. The **Forecast5Hour** vintage is issued approximately five hours before its target and is therefore genuinely available at decision time across the whole horizon.

Every model is trained on a chronological split and reported against a persistence baseline, and the renewables model additionally against the raw Energinet forecast it post-processes.

3.3 III. Genetic optimization of a dispatch trajectory

For the forecasted horizon, the optimal combination of sources is inferred using genetic algorithms. In order to escape eventual local optima and increase the method’s success, this is done by employing a genetic ensemble with 16 optimizers. Populations of solutions are initialized stochastically and transformed by evolutionary processes to arrive at the “fittest” distribution. The restrictions are the ranges of power plant production and the interconnector limits for exchanges, inferred from the past year’s data and capped by technical transfer capacity. Ramp limits are taken from the 99th percentile of realised 15-minute changes over the same year, which describes what the fleet can actually do better than any fixed assumption. At 1 MW resolution a single time step admits approximately 8×10^{34} combinations.

A chromosome is a trajectory. A chromosome is the whole four-hour path rather than a single time point. Writing $d = 11$ for the number of genes per time step and $T = 16$ for the horizon, each individual is a matrix $X^{(j)} \in \mathbb{R}^{T \times d}$ whose rows $x_t^{(j)}$ are the distributions at each step, and the first generation is

$$G = \begin{bmatrix} - & \text{vec}(X^{(1)}) & - \\ - & \text{vec}(X^{(2)}) & - \\ & \vdots & \\ - & \text{vec}(X^{(m)}) & - \end{bmatrix} \in \mathbb{R}^{m \times Td}, \quad (3)$$

where m is the size of the population. Coherence is imposed directly, as a ramp constraint between consecutive rows.

The fitness function. Each chromosome is evaluated using the regression ensemble to infer the CO₂ emission level implied by the solution it represents, while solutions that do not balance the power grid, violate a ramp limit, leave the region where the model has evidence, or propose an unprecedented outcome are penalized. Each optimal trajectory minimizes

$$X^* = \underset{X^{(j)}}{\operatorname{argmin}} \frac{1}{T} \sum_{t=1}^T \left[\xi_b \mathcal{P}_b(x_t) + \xi_r \mathcal{P}_r(x_t, x_{t-1}) + \xi_s \mathcal{P}_s(x_t) + \xi_f \mathcal{P}_f(x_t) + \zeta \Theta(x_t) + \gamma \sigma(x_t) \right], \quad (4)$$

whose terms are, in order,

- the **grid imbalance penalty**

$$\mathcal{P}_b(x_t) = \mathbf{I} \left(\left| D_{XG,t} - R_{XG,t} - \sum_{i \in \mathcal{B}} x_{t,i} \right| > \tau D_{XG,t} \right) \left(1 + \frac{|D_{XG,t} - R_{XG,t} - \sum_{i \in \mathcal{B}} x_{t,i}|}{\tau D_{XG,t}} \right), \quad (5)$$

where $\mathbf{I}$ is the indicator function, D_{XG} is the demand forecast and R_{XG} the renewable forecast provided by the XGBoost regressors, and $\tau = 0.01$. Scaling the indicator by the relative size of the violation gives the search a gradient towards feasibility while preserving the hard threshold;

- the **ramp penalty** $\mathcal{P}_r(x_t, x_{t-1}) = \sum_i \max(0, |x_{t,i} - x_{t-1,i}| - \rho_i) / \rho_i$, with ρ_i the per-resource ramp limit and x_0 the latest observed distribution;
- the **support penalty** $\mathcal{P}_s(x_t) = \mathbf{I}(n(x_t) < n_{\min}) + \max(0, d^2(x_t)/d_q^2 - 1)$, which confines the search to the region where the emissions model has evidence, with $n_{\min} = 20$ and d_q^2 the 99th percentile of the training Mahalanobis distances;
- the **plausibility-floor penalty** $\mathcal{P}_f(x_t) = \mathbf{I}(\Theta(x_t) < \phi_t) (1 + (\phi_t - \Theta(x_t)) / \max(\phi_t, 1))$, where ϕ_t is the 5th percentile of metered emission rates among training hours whose demand and renewables lie within 200 MW of the forecast for step t (widened to 400 MW and then 800 MW if fewer than 25 matches are found). This term asks whether the *outcome* has precedent, which a distance in feature space does not;
- the **emissions level** Θ inferred by the regression ensemble, augmented by ζ ; and
- the **model dispersion** σ , augmented by γ , which makes the optimizer prefer solutions the model is confident about.

Feasibility by construction. Every candidate is passed through a repair operator before evaluation. Walking the horizon forward, the admissible box at each step is the intersection of the technical limits with the ramp window around the previous step; the residual energy imbalance is then distributed across the balance-carrying resources in proportion to their remaining headroom. This turns feasibility from something the search must rediscover into a property of the representation, so the evolutionary effort is spent on emissions instead.

Implementation. The ensemble of independent optimizers is evaluated as a single batched array, an island model with periodic migration of the best individual between islands. Emissions, dispersion, leaf support and novelty all come out of one traversal per generation. With 16 islands, populations of 150 and 200 generations — 4.8×10^5 evaluations — a full solve takes approximately 80 seconds, which fits comfortably inside a scheduled job refreshing every 30 minutes.

4 Results

4.1 The emissions model

On a chronological hold-out of 6,920 observations (June–September 2026) the monotone ensemble achieves an intensity MAE of 11.05 g/kWh and a total emission-rate R^2 of 0.889. Holding every other resource fixed and sweeping large-plant output from 0 to 1,500 MW, predicted total emissions rise by 384 t/h and the minimum sits at zero, which is the relationship the monotonicity constraint is there to enforce.

4.2 The forecasts

Figure 1 reports forecast error against horizon on a chronological hold-out. Demand RMSE grows from 88.9 MW at 15 minutes to 374.5 MW at four hours, against a persistence baseline of 98.3 MW and 815.8 MW. Renewables RMSE grows from 135.9 MW to 468.6 MW, against 151.7 MW and 1394.0 MW for persistence and a flat 699.5 MW for the raw Energinet forecast; post-processing that forecast with recent observations improves on it at every horizon.

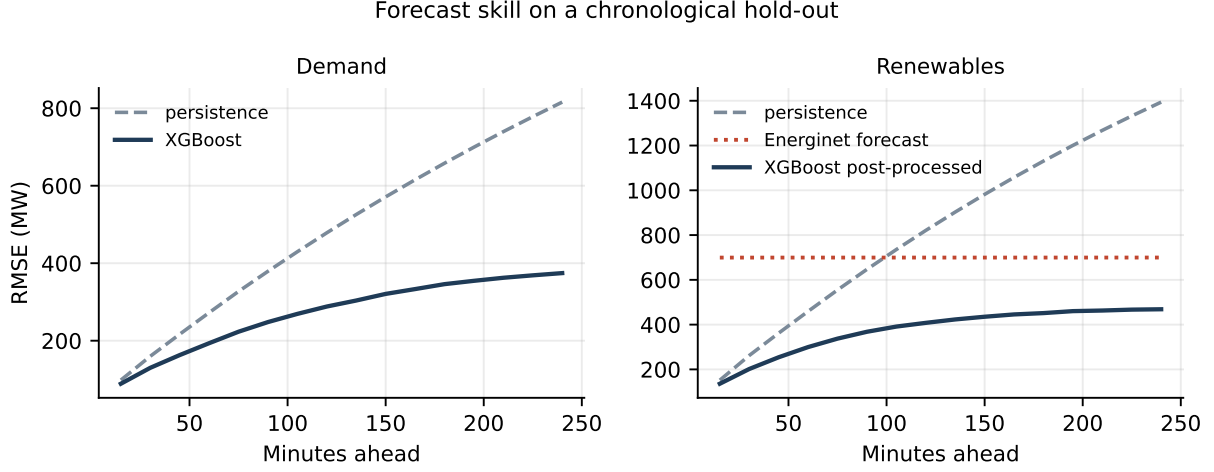


Figure 1: Forecast skill against horizon, with persistence and, for renewables, the raw Energinet forecast as baselines.

4.3 The optimizer

An evolutionary heuristic is only worth its cost if it beats random search under the same evaluation budget. Figure 2 shows that it does: at 4.8×10^5 evaluations and matched wall-clock time, the genetic ensemble reaches a mean intensity of 60.8 g/kWh against 87.7 g/kWh for random search, and does so while keeping every step inside the region where the emissions model has support.

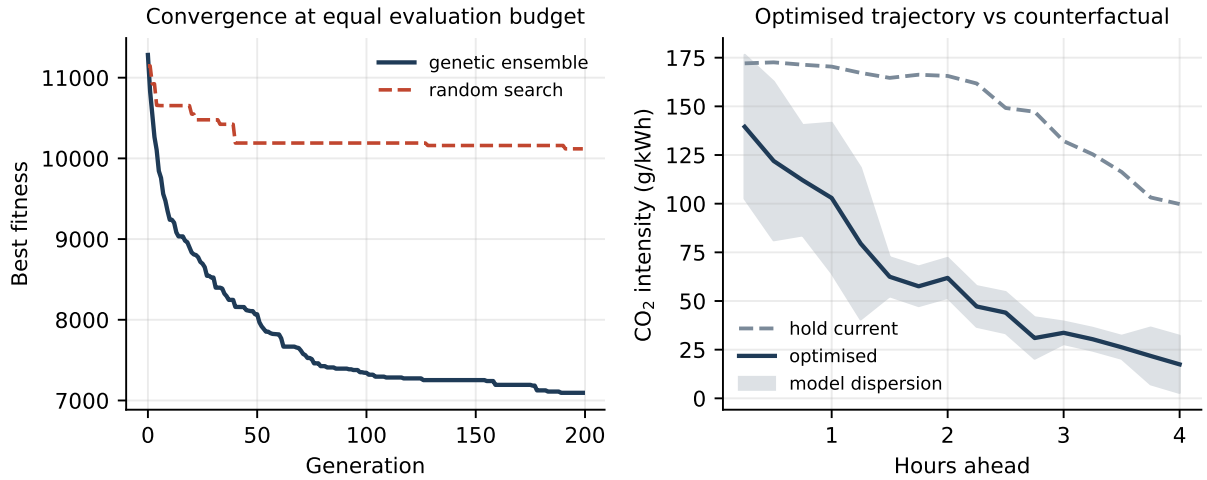


Figure 2: Left: convergence against random search at equal evaluation budget. Right: the optimized intensity trajectory against the hold-current counterfactual, with the emissions model’s own dispersion shaded.

4.4 How the reduction is measured

The reduction reported on the live dashboard is measured against a hold-current counterfactual, in which demand and renewables follow their forecasts but the controllable resources stay where they are. Comparing against the present measured value instead would credit the optimizer with whatever the weather was going to do anyway: if wind is rising over the horizon, intensity falls without any intervention at all.

On the representative solve shown in Figure 3, the present measured intensity was 204.1 g/kWh,

the counterfactual averaged 140.0 g/kWh over the horizon, and the optimized trajectory averaged 63.8 g/kWh. The reduction attributable to re-dispatch on that afternoon is therefore 54.4%. That number is a single solve. The distribution across seasons is the subject of Section 5.

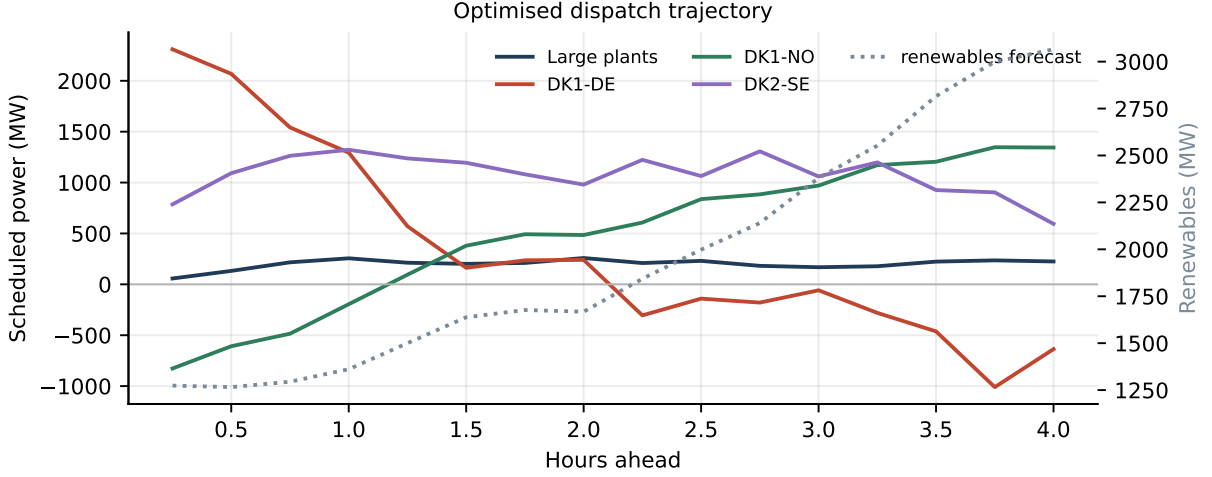


Figure 3: An optimized dispatch trajectory. As forecast wind rises through the afternoon the schedule withdraws from German imports, crossing into export, and shifts towards Norwegian hydro, while thermal output stays low. Every change respects its ramp limit.

Every step of that trajectory satisfies the energy balance to within a fraction of a megawatt and every change respects its ramp limit; the maximum balance error over the horizon is 2.4×10^{-4} MW.

5 Historical backtest

A single afternoon does not establish that the system works. The live pipeline reports what the emissions model says about a schedule the model itself helped to choose. The backtest asks whether those claims survive contact with the dispatch that actually ran, under models that have never seen the period being scored.

The year is split into five expanding windows. Cutoffs fall on 4 February, 21 March, 6 May, 21 June and 6 August 2026. Within each fold the emissions ensemble, both forecasters and the operating envelope are fitted only on data preceding the cutoff; hyperparameters are tuned once on the earliest window and reused, so no fold sees its own future. Forty-five origins are drawn at random from each test window, restricted to spans that sit on an unbroken 15-minute record. That yields 225 origins and 3,600 horizon steps between February and September. Each origin is replayed at the deployed genetic-algorithm budget.

5.1 The emissions model against the meters

On the dispatch that actually happened — the only place in the exercise where a ground truth exists — the out-of-sample model achieves $R^2 = 0.941$, a mean absolute error of 49 t/h, and a signed bias of -14 t/h. The median relative error is 12%.

That is a test of levels. The optimizer never asks the model for a level; it asks which of two schedules is cleaner. Figure 4 tests exactly that comparison, on pairs of historical moments matched to within 75 MW of demand and of renewables. The balance identity then forces both members of a pair to serve the same load, so what differs is purely how it was served. Both members have a metered outcome.

Under matched demand and wind, does a predicted saving actually arrive?

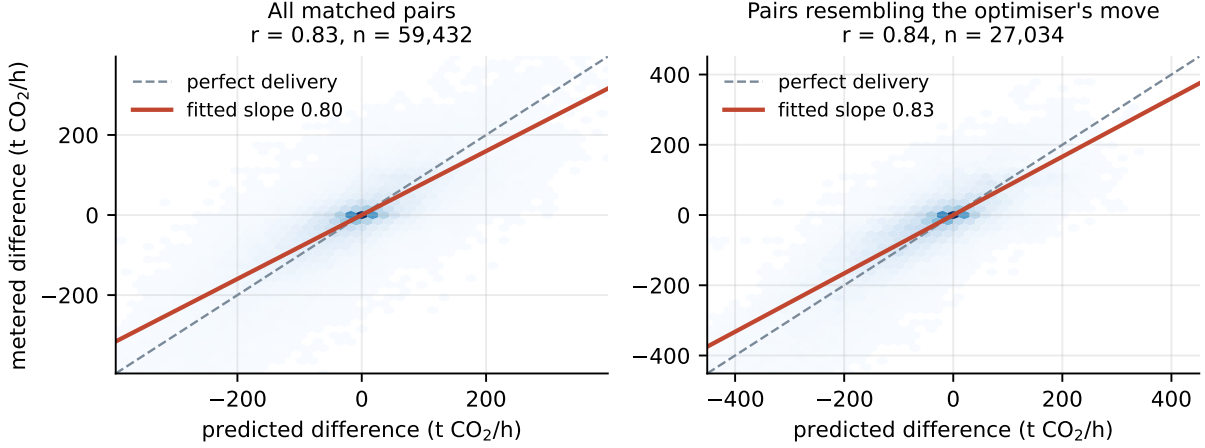


Figure 4: Predicted versus metered difference in the total emission rate, for pairs of hours matched on demand and wind. A slope of 1 would mean a predicted saving of X t/h is delivered in full. Left: all 59,432 pairs. Right: the 27,034 pairs whose dispatch difference points in the same direction as the optimizer’s own move.

When the model says one schedule is X t/h cleaner, the meters deliver $0.80X$ on average ($r = 0.83$, sign agreement 78%). Restricted to pairs within seven days, which controls for seasonality, the slope is 0.76. Restricted to pairs that resemble the optimizer’s characteristic move — less domestic thermal, more low-carbon import — the slope is 0.83 ($r = 0.84$). A linear model of empirical marginal emission factors, fitted on the same training windows, ranks the same pairs far worse ($r = 0.58$). The constrained ensemble is therefore earning its complexity, and the slope is measured over a range the grid has actually explored: the optimizer’s median L1 move is 3,863 MW, and 56% of the observed pairs differ by at least that much.

Matching is never perfect. Two moments with identical demand and wind can still differ in what the neighbouring grids were burning, which the feature set cannot see. The recovered slope is therefore an upper bound on how well any model of these features could rank dispatches.

5.2 What the optimizer would have claimed

Pooled over all 3,600 horizon steps, the pipeline reports a 35.1% reduction against the hold-current counterfactual and a 35.8% reduction against the dispatch that actually ran. After the two guards of Section 3.1 — softplus on the ensemble output and the plausibility floor in the fitness — no proposed step has a negative predicted emission rate, and 99.9% of steps sit at or above the historical floor computed from the forecast conditions of that origin.

Applying the ranking slope of 0.83, a predicted tonne is 0.83 of a delivered tonne, which takes the 35.8% to 29.7%. Capping each proposal at the 5th percentile of historically observed outcomes under comparable demand and wind, then applying the same discount, gives 27.2%. That is the figure the evidence supports. It is stable across the year: the pooled reduction against realised dispatch is between 31% and 39% in every fold.

5.3 Where the reduction comes from

Of the 129 t/h the model attributes to re-dispatch, 109 t/h (85%) comes from changing the cross-border schedule and 17 t/h (13%) from running less domestic thermal generation. Over a four-hour horizon the optimizer moves Danish generation by a median of 10 MWh and net

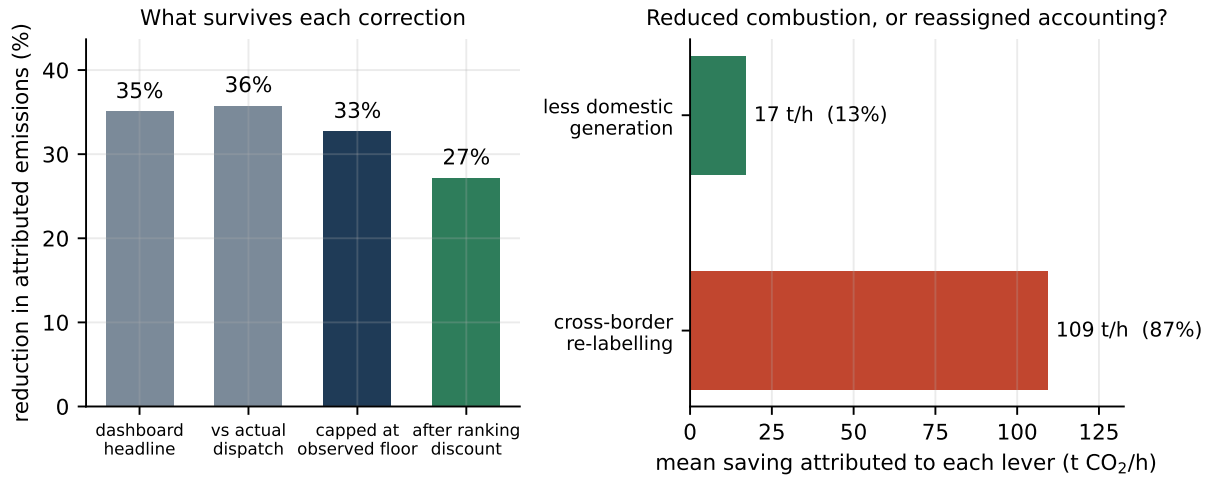


Figure 5: Left: the reduction after each correction that the backtest can apply. Right: the share of the remaining saving that comes from running less domestic generation versus from changing the cross-border schedule.

imports by a median of 389 MWh. Under consumption-based accounting that is a real reduction in Denmark's attributed intensity. Whether it is a reduction in atmospheric CO₂ depends on the Nordic and German marginal plants, which this system does not observe.

5.4 Feasibility once the forecast is replaced by what happened

The genetic algorithm balances the schedule against its own forecast. Under the demand and renewables that materialised, the mean absolute imbalance is 87 MW (1.9% of demand) at 15 minutes and 489 MW (10.5%) at four hours. Only 13% of individual steps remain inside the 1% tolerance the optimizer enforces. The residual is the forecast error — 221 MW MAE for demand and 286 MW MAE for renewables, averaged over the horizon — which a real operator covers with regulating reserve. That plant is typically the fastest and dirtiest on the system, and its emissions cost is not in the fitness function.

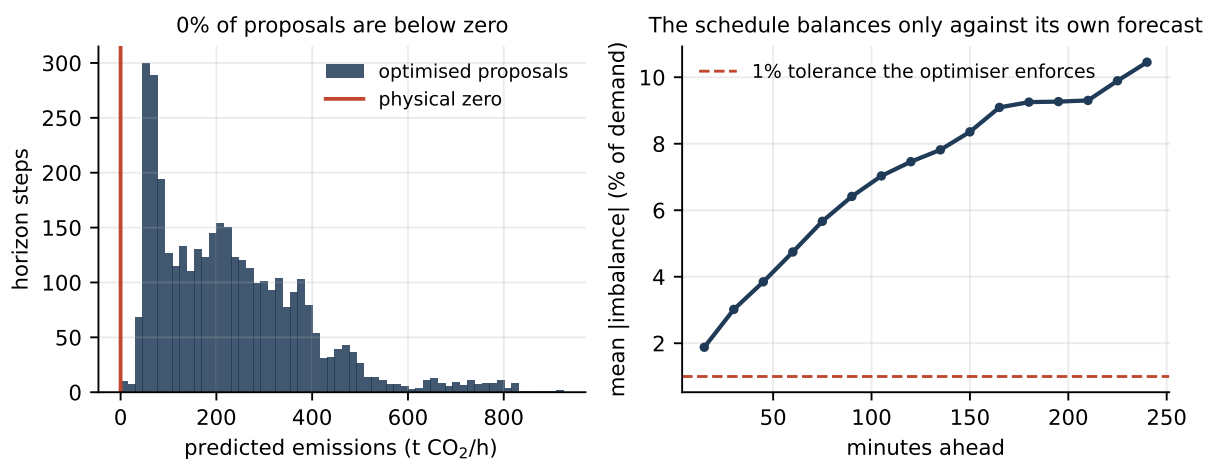


Figure 6: Left: the distribution of predicted emission rates on optimized proposals; the physical zero is marked. Right: mean absolute imbalance of the optimized schedule under realised conditions, against the 1% tolerance the optimizer enforces on its own forecast.

The Mahalanobis novelty guard, by contrast, barely discriminates. Optimized points have a *lower* median novelty (12.2) than realised ones (14.7), and 98% of both pass. Model error on

realised points does grow with novelty, from 38 t/h in the lowest quintile to 65 t/h in the highest, so the signal exists; a distance on the marginal feature distribution does not fire on the proposals that matter. The plausibility floor is what closed that route.

6 Limitations

The framework does not model unit commitment. Large-plant output is treated as a single aggregate quantity, where in reality it is dozens of units with individual minimum stable loads and start-up times.

Interconnector limits are fixed over the horizon. Real transfer capacities are published per market time unit and are frequently derated; a production system would consume them rather than assume them.

The model takes no account of exchange contract terms or energy prices, so a proposed schedule may be physically feasible and economically impossible. Cross-border flows are in any case outcomes of coupled European market clearing, not dials a single operator turns. The backtest measures what the emissions model says about a hypothetical re-dispatch; it does not establish that any actor could have executed it.

The emissions model is compressive, as a regularized estimator fitted to a long-tailed target will be. Absolute emission levels should be read as estimates with roughly a 10 g/kWh error bar, not as measurements. The *reduction* is more robust than the levels, because the counterfactual and the optimized trajectory are evaluated with the same model; the ranking test of Section 5 is the quantity that converts that shared evaluation into evidence.

Most importantly, Energinet’s CO₂ figure is a *consumption-based* intensity for Denmark: it assigns emission factors to imports. A schedule can therefore lower the Danish number by importing more low-carbon power from Norway or Sweden without lowering emissions anywhere. Nordic hydro export capacity is finite, so power drawn to Denmark is power not delivered elsewhere, and the displaced consumer may burn gas instead. The reductions reported here are reductions in Denmark’s attributed intensity. Establishing how much of that is genuine abatement requires marginal emissions data for the surrounding system, which is outside the scope of this work.

7 Future work

Three directions follow from the limitations above. Replacing average with marginal emission factors would let the objective express abatement rather than attribution. Restricting the decision variables to resources an identified actor actually controls would turn the output from a hypothetical into a recommendation. And introducing semi-continuous constraints for minimum stable load would turn the problem into a genuine mixed-integer program, where the choice of an evolutionary heuristic could be benchmarked against an exact solver.

Visit the web app, which includes a dynamic comparison between recent emissions in Denmark’s power system, the hold-current counterfactual, and the minimized CO₂ levels over the next four hours, as well as the optimal energy resource distributions leading to these emission levels and the historical backtest summarised in Section 5.